Wrapture: AI-Powered Eco-Packaging for Greener E-Commerce

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***Abstract*— Oversized or poorly matched packaging introduces material, shipping volume, and damage risk in e-commerce, but many small sellers still choose packaging by hand. This paper presents Wrapture, a web-based decision-support system that chains product detection, catalogue lookup, materials classification, dimensional packaging fit, ranking, 3D visualization, and sustainability information in one flow. YOLO11 detects product category in the image, and category is used to look up dimensions and weight from Amazon Berkeley Objects (ABO) records. A deterministic fit engine then tries product orientations against 2,209 packaging records with at least 10% clearance and ranks the packages that fit by clearance, space utilization, material, protection, fragility, price, plus available environmental data. Of 42,651 product records evaluated, 40,200 had at least one fitting package, a 94.25% dimension-based packaging-fit coverage; this measures catalogue coverage and is not a recommendation accuracy. An independent MobileNetV2 classifier, trained on 2,616 images and tested on 645 unseen images, achieved 86.98% accuracy over seven material classes and runs in the browser via TensorFlow.js. The system is delivered as a React app plus a packaging catalogue, 3D simulator, and an administrative dashboard.**

Keywords*— *E-commerce packaging, sustainable packaging, YOLO11, MobileNetV2, packaging optimization, computer vision, packaging recommendation, 3D visualization.

I. Introduction

Product packaging that is too big is wasteful, takes up space, and often requires void fill, while packaging that is too small exposes the product to damage. Sellers with broad product catalogs, and small sellers in particular, rarely have the luxury of a packaging engineer, and default to heuristics based on a small set of familiar sizes. Human intuition lacks the ability to trade-off between item dimension and fragility, material and price, across thousands of potential packaging options. The solution lies in recommendation systems that operate on product and packaging catalogs, with features tied to specific products, and that can provide contextual explanations. This paper introduces Wrapture, a web application that brings together product detection, catalog look-up, material classification, dimensional fit, recommendation ranking, catalog browsing, 3D visualization, and sustainability information into a single application. Wrapture uses 42,651 product records from the Amazon Berkeley Objects (ABO) [1] catalog, and a 2,209-record packaging catalog derived from Packrift data [2]. A YOLO11 detector [3] identifies the category and bounding box of the product in an image query. Using the category, the product's dimensions are retrieved from ABO [1], with the note that the category detection serves as a reference, and not as an identifier of a commercial SKU. The retrieved dimensions are passed to a deterministic fit engine, that selects applicable options from the packaging catalog that meet a 10% clearance threshold, and ranks them with a relevance score. Finally, a MobileNetV2 classifier [4] is used to identify the material of an image in a separate support task.

The contributions of this paper are:

- an end-to-end pipeline from category-level YOLO11 detection [3] into ABO catalog look-up and packaging fit estimation with material classification serving as an independent support task;
- a deterministic, clearance-aware fit and ranking algorithm that utilizes dimension-based coverage statistics across 42,651 product records and 2,209 packaging records;
- a seven-class MobileNetV2-based material classification model [4] with 86.98% test accuracy, deployed in the browser via TensorFlow.js; and
- a React-based application with a catalogue browser, 3D simulation and visualization, and an administrative panel.

II. Related Work

A. Vision Models for Product and Material Recognition

MobileNetV2 utilizes inverted residual blocks with linear bottlenecks to maintain a low level of computation [4], which is appropriate for classification on the client-side, while TensorFlow.js enables the execution of such a model within a web browser [5]. The YOLO family of detectors is able to predict classes and bounding boxes of objects in a single forward pass [6], while YOLO11 is the Ultralytics release used in this paper [3]. A detector of this kind predicts a category label, an associated box, and a confidence score. Many products of various sizes and brands fall under the same category, and thus the output does not distinguish a commercial SKU. Wrapture, therefore, decouples three tasks that a single model cannot tackle reliably: detecting the product category, obtaining catalogue information, and classifying visual material.

B. Packaging Optimization

The three-dimensional bin packing problem is to find an orthogonal arrangement of many rectangular items in the fewest bins and is strongly NP-hard [7]. The selection problem in Wrapture is narrower: one product is matched against a fixed catalogue of package sizes. The deciding questions are whether the product fits in some orientation with enough clearance, and

how the fitting packages compare. A single-orientation comparison can reject a package that would accept the product once it is rotated. Wrapture tests every orientation of each product-package pair and ranks the survivors on several criteria instead of volume alone.

C. Sustainable Packaging

The sustainability of packaging is dictated by the materials used, the size of the package, and the end-of-life processing. The U.S. EPA Waste Reduction Model (WARM) provides greenhouse-gas emission factors for material-management scenarios such as recycling and landfilling [8] and its containers and packaging sections include paper, plastics, metals, and glass [9]. Estimates based on these factors therefore, require reliable information about the mass of a package and its material composition. Weight fields in a packaging catalogue typically define a multipack and not the package itself, and so Wrapture only displays the environmental figures if they are provided in the catalogue and does not infer package-level values from ambiguous data

D. Research Gap

Product recognition, material classification, and packaging optimization are typically regarded as separate problems. A useful recommendation tool must combine them into one workflow that is understandable and grounded in verifiable data. Wrapture combines product detection, catalogue lookup, geometrical fit, ranking, material classification, and 3D visualization in one application.

III. Proposed System

A. System Overview

Wrapture is a web-based packaging recommendation system implemented in the form of a React application, while the service, providing catalogue and recommendation data, is based on Flask. Fig. 1 represents the system architecture, which consists of the product data module (ABO), the packaging catalogue module (based on the Packrift-derived data), data validation, YOLO11-based product detection, 3D fit engine, recommendation and ranking module, 3D packaging simulator, sustainability, and the material classification module (MobileNetV2).

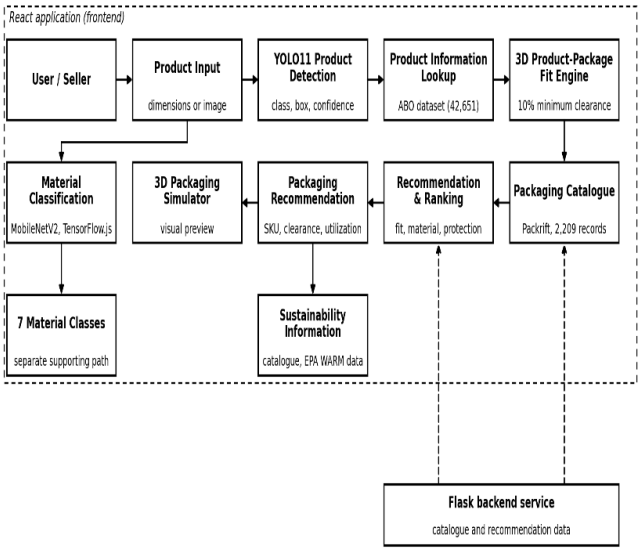


Fig. 1. *Architecture of the Wrapture system. Material classification runs as a separate supporting path.*

A new path begins with user-entered manual dimensions or an image of the product. The latter, in the case of YOLO11, yields a category, a bounding box, and a confidence level. The category is matched against the ABO catalogue to find an entry with compatible dimensions and, if found, provides a weight, otherwise, the weight is manually entered. The resulting, validated dimensions and weight are then passed to the fit engine, which narrows down the catalogue entries to those that can physically fit and calculates clearance and utilization levels, ranking the options by these criteria. The recommended packaging can then be visualized in the 3D simulator, and sustainability information relating to it is made available to the user. Material classification is a completely separate path: the MobileNetV2 model classifies images into one of seven material categories and has no bearing on the package recommendation process.

B. Product Dataset

ABO [1] is the source of product information, providing identifiers, names, product types, dimensions, and weight if available. The processed catalogue contains 42,651 product records. Product volume is

$$V_p = L_p W_p H_p \tag{1}$$

where L_p, W_p, and H_p are the product length, width, and height. Dimensions are validated prior to a record being passed into the fit engine.

C. Packaging Dataset

The packaging catalogue was extracted from the Packrift Cube Efficiency Atlas [2] and contains 2,209 entries of boxes, mailers, and poly bags each with an SKU, name, packaging family, product type, length, width, height, volume, material, and protection information with price and weight where available. The package volume was recalculated as

$$V_k = L_k W_k H_k \tag{2}$$

and compared to the volume reported in the catalog during validation.

D. Preprocessing and Validation

The fit decisions are based only on the dimensions. Therefore, the records need to be validated before their use. First, product and package dimensions are checked, volumes are computed, material and protection fields are filled, and records without sufficient data for the fit calculation are removed. The catalogue is then converted to JSON for use by the web application. A material label is assigned to a record only if the catalogue data supports it; otherwise, the field is marked as unknown.

E. Packaging Material and Protection Information

The materials of the package can be categorized into corrugated, plastic, paper, and unknown. Information on protection is retrieved based on the fields available in the catalogue, such as the type of wall, ECT rating, and heavy-duty indication. The system does not include generic protection ratings that cannot be validated; instead, it stores the data from the catalogue and applies it directly to the ranking process.

IV. Packaging Recommendation Method

A. Clearance-Aware Multi-Orientation Fitting

Every product-package pair is tested in all orientations, that is, all permutations of the three product dimensions. A package is physically compatible if at least one orientation satisfies

$$L_p(1 + c) \le L_k \tag{3}$$

$$W_p(1 + c) \le W_k \tag{4}$$

$$H_p(1 + c) \le H_k \tag{5}$$

where c is the required clearance fraction. The implementation uses c = 0.10.

B. Clearance and Space Utilization

For a given orientation, the clearance along each axis is

$$C_L = (L_k - L_p) / L_p \tag{6}$$

$$C_W = (W_k - W_p) / W_p \tag{7}$$

$$C_H = (H_k - H_p) / H_p \tag{8}$$

and the minimum clearance is

$$C_{min} = \min(C_L, C_W, C_H) \tag{9}$$

Conditions (3)–(5) hold exactly when $C_{min} \ge c$, so a package is not accepted because one dimension is generous while another is tight. Space utilization and unused volume are

$$U = (V_p / V_k) \times 100 \tag{10}$$

$$V_u = V_k - V_p \tag{11}$$

A higher U indicates a dimensionally tighter package, and V_u is the volume left unoccupied.

C. Deterministic Ranking

Packages that passed the fit test are used as a candidate set and sorted based on the physical fit, clearance, space utilization, empty space, package material, protection, product fragility, price (if available), and environmental information (if available). This ranking is a result of a rule-based engineering model with hand-chosen weights. The model is not a trained recommendation system, and no claims are made about its recommendation accuracy.

Product fragility changes the weight distribution of the criteria to prioritize the protection and, hence, the material and, to a lesser extent, the space utilization for fragile and moderately fragile items and prioritize space utilization for non-fragile items. In case of a physically fitting package, but without any evidence of protection in the catalogue for fragile items, such a recommendation would not be presented as fully validated. All candidates should be presented with their SKU, dimensions, material, protection, clearance, utilization, and price to give the user an insight into the reason for their ranking.

V. Material Classification Model

A. Dataset

The model was trained on the Rootstrap Waste Classifier dataset [10], which contains 7 categories of waste: cardboard, compost, glass, metal, paper, plastic, and trash. The number of images in the training split is 2,616, and the test set consists of 645 unseen images. According to the dataset's card, the images were collected from the Internet with white backgrounds [10].

B. MobileNetV2 Architecture

MobileNetV2 [4] pretrained on ImageNet [11] is used as a frozen feature extractor followed by global average pooling, dropout, dense output layer, and softmax. By freezing the backbone, only a few parameters are subject to training [12]. The input image is resized to

$$224 \times 224 \times 3 \tag{12}$$

The total number of parameters in MobileNetV2 is 2,266,951, of which only 8,967 are trainable. The number of trainable parameters coincides with a fully connected layer with 1,280 inputs and 7 outputs (1,280 × 7 weights plus 7 biases). The model was trained for 10 epochs.

C. TensorFlow.js Deployment

The trained model is exported in the TensorFlow.js format [5] for inference in the browser. The deployed files consist of a model configuration, weight shards, and a class-name list. The React scanner loads the model via the TensorFlow.js library, resizes the input image to 224 × 224, and scales pixel values with

$$x' = x / 127.5 - 1 \tag{13}$$

The model predicts a probability for each of the seven classes, and the application displays the class with the highest probability.

VI. Implementation

A. React Application

Wrapture is a React application backed by a Flask-based service that supplies catalogue and recommendation data. Its bottom navigation contains Home, Scan, Packages, Simulator, and More. The All Tools view lists the Packaging Recommender, AI Product Scanner, 3D Packaging Simulator, Sustainability Insights, and EcoShop packaging catalogue. TrashGPT packaging chat, EcoPoints, Returns, Reviews, and Reverse Logistics also appear as interface modules; they lie outside the evaluation reported here. A user moves from product input, through detection and material analysis, to a ranked recommendation with alternatives, a simulator preview, and available sustainability information.

B. YOLO11 Product Detection

The AI Product Scanner runs the YOLO11 detector [3] on a captured or uploaded product image and returns a category label, a bounding box, and a confidence score for each detection (Fig. 2(a)). When several objects are detected, a single detection is selected, and its category is passed to catalogue retrieval. The detector identifies a product category, not a commercial SKU.

C. Product Information Retrieval

The selected category is used to interrogate the ABO catalogue for a matching dimension and weight record. If such a record exists then its dimensions are passed to the fit engine. If not then the dimensions are input manually. The decision on the packaging is thus based upon catalogue data rather than an estimated size based upon image pixels.

D. Material Classification Tool

The Material Analysis tool applies the classifier of Section V to a captured or uploaded image, returning one of the seven classes (Fig. 2(b)). It runs independently of product detection and packaging recommendation.

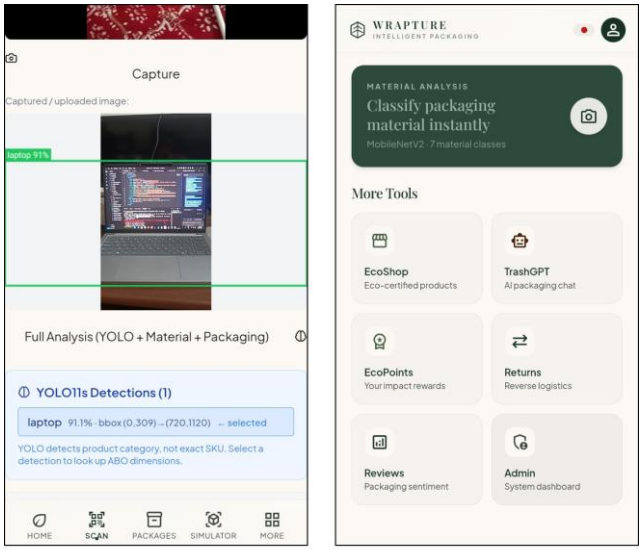


Fig. 2. *Vision workflow screens: (a) YOLO11 product detection with a selectable detection; (b) Material Analysis entry point (MobileNetV2, seven classes*).

E. Packaging Recommendation Engine

The method of Section IV is exposed through the Packaging Recommender, which accepts validated product dimensions and weight and returns ranked packaging options. Each candidate carries its clearance, utilization, material, and protection information.

F. Packaging Catalogue

The EcoShop interface contains all 2,209 packaging records and allows searching by title, SKU, or material, with filters (Fig. 3(a)). Each entry includes the dimensions of the package, its unit price, the number of items, the type of protection, and, if available, a recycling CO2-equivalent figure per unit based on EPA WARM, with a link to the source record in Packrift.

G. 3D Packaging Simulator

The simulator takes a product type, size, weight and a choice of materials for packaging (corrugated cardboard, kraft paper, plastic/poly or best-fit) as inputs and then displays a 3-D visualization of the product in the chosen packaging based on catalog data (Fig. 3(b)), indicating explicitly that the output is only a visualization and not a CAD-scaled or physically accurate depiction.

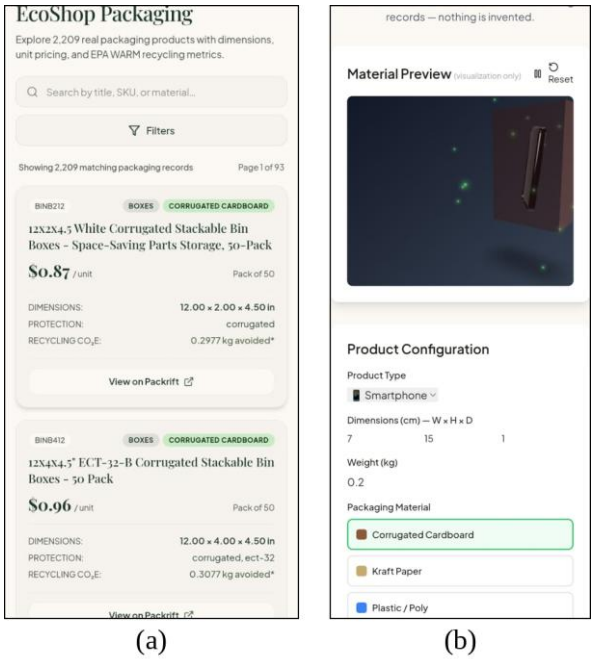


Fig. 3. *Catalogue and simulator screens: (a) EcoShop packaging catalogue with search, filters, and per-listing data; (b) 3D packaging simulator with product configuration*.

H. Sustainability Analysis

The Sustainability Insights tool provides volume- and CO2-related information based on the data from the packaging catalogue and EPA WARM [8], [9]. The figures are derived from the catalogue and thus represent merely catalog-level conclusions and not a lifecycle analysis. They are available depending on the data provided by the source. The EcoPoints and review panels offer this information to users as an informational and entertainment feature and not as a reliable analytical tool.

I. Administrative Dashboard

A separate administrative dashboard displays panels with the information on the number of total scans, products found, recommendations generated, packaging options provided, packaging activity, recent activity, and status. The panels describe what the system interface looks like; no information on the backend utilization or performance is given in the paper.

VII. Results and Discussion

A. Material Classification Results

Table I summarizes this result. Overall accuracy of the classifier was 86.98% over the 645 unseen test images (561 predicted correctly). This score is based on the Rootstrap test set. In this case, it reflects the model's ability to distinguish these 7 material

classes on this specific test set. However, it makes neither statements about the packaging choice nor about the performance on mobile phone camera images.

TABLE I. MATERIAL CLASSIFICATION RESULTS

Metric	Result
Model	MobileNetV2
Dataset	Rootstrap Waste Classifier
Training images	2,616
Test images	645
Number of classes	7
Input size	224 × 224 × 3
Training epochs	10
Test accuracy	86.98%
Total parameters	2,266,951
Trainable parameters	8,967

B. Packaging Catalogue Coverage

Table II reports how many of the 42,651 ABO product records have at least one catalogue package that satisfies the fit condition of Section IV under c = 0.10. Coverage is the number of products with at least one fitting package divided by the number of products evaluated:

$$\text{Coverage} = (40{,}200 / 42{,}651) \times 100 = 94.25\% \qquad (14)$$

This is dimension-based packaging-fit coverage: the share of evaluated products for which at least one of the 2,209 catalogue packages passes the physical fit test. It is not recommendation accuracy, prediction accuracy, or model accuracy, because the ranking is rule-based and no outcome labels exist to score it against. The remaining 2,451 products (5.75%) have no dimensionally compatible package in the current catalogue.

TABLE II. PACKAGING CATALOGUE COVERAGE

Metric	Result
Product records evaluated	42,651
Packaging options	2,209
Products with suitable packaging	40,200
Products without suitable packaging	2,451
Dimension-based coverage	94.25%

C. Representative Recommendation Results

Table III lists representative ABO records processed through the engine. For a given category, the engine returns several possible candidate packages, and not a specific type for a category. In the drawer-slide record, a package of around 2 × 2 × 25 in provides a physical fit with approximately 13.64% of minimum clearance. Observed utilization is determined by a product's shape and available packing: 17.94% for drawer slides, 32.11% for work gloves, 50-62.5% for a pet bed, 68-74% for a dining chair, and 83.79% for a flat sheet product. The large garment rack demonstrates the importance of fragility: a physical fit exists, but a proof of protection for a fragile product can be lacking.

TABLE III. REPRESENTATIVE RECOMMENDATION RESULTS

Product	Dimensions (in)	Recommendation observation
Drawer slides	22 × 0.87 × 0.9	Suitable compact packaging identified
Folding stair climber cart	19.5 × 16.25 × 43.87	Large packaging identified
Basil product	1.9 × 1.95 × 4.15	Multiple compact packages identified
Outdoor throw pillow	24 × 12 × 4	Multiple fitting packages identified
Large garment rack	19.49 × 35.43 × 59.06	Physical fit exists, but protection for a fragile product may be insufficient

D. Functional Testing

Table IV lists the functions of the implemented application; each was tested and worked as expected. These are tests of the functionality, not performance benchmarks. For YOLO11 no accuracy at detector level is reported (Fig. 2(a) is only one example), and no user study is reported.

TABLE IV. FUNCTIONAL TESTING

Test case	Expected result	Status
Product image input	Image accepted	Pass
YOLO11 detection	Product detected	Pass
Product information lookup	Available ABO information retrieved	Pass
Material classification	Material category predicted	Pass
Packaging catalogue	2,209 records loaded	Pass
Physical fit	Suitable packages identified	Pass
Recommendation	Ranked alternatives displayed	Pass
3D simulation	Packaging visualized	Pass
Sustainability information	Available information displayed	Pass

E. Discussion

The results present two distinct components. Fit coverage is limited by the catalogue; thus, expanding or enhancing the 2,209 packaging records is the most straightforward method to decrease the 5.75% of products for which there is no match. The 86.98% material accuracy was achieved in a supporting classifier that does not select packages. Since the pipeline attaches a detection to a catalogue record, and not to a size estimated from pixels, a detection error alters which record is retrieved, keeping the fit calculation as a deterministic function of retrieved dimensions. The environmental figures shown alongside packages are catalogue and EPA WARM indicators; this paper reports no measured carbon or cost savings.

VIII. LIMITATIONS

YOLO11 detects the product category and location but cannot guarantee the SKU, and a single detection is selected. Catalogue matching relies on structured ABO data, and some product dimensions require manual entry. The 86.98% classifier accuracy was attained on the Rootstrap test split and is dependent on that data, and results on images captured with a mobile camera and varied backgrounds and lighting might differ. Packaging covers only the 2,209 available records, leaving 2,451 products without any package suitable. Recommendation ranking is deterministic and has no learning

from past seller outcomes. Catalogue price and weight fields might denote multipacks, so per-package cost and environmental values cannot be determined precisely, and sustainability values rely on the quality of the catalogue data. Automatic IoT weighing is not built in. The 3D simulator is a visualization and is not a CAD-grade engineering simulator.

IX. Conclusion and Future Work

A. Conclusion

This paper introduced Wrapture, a packaging decision-support system, which linked YOLO11 category detection, ABO catalogue lookup, deterministic multi-orientation packaging fit, and ranking, along with a separate MobileNetV2 material classifier, a packaging catalogue, a 3D simulator, and sustainability information in a React application. Against 42,651 product records and 2,209 packaging records, the fit engine found at least one package with 10% clearance for 40,200 products, a dimension-based coverage of 94.25%. The material classifier reached 86.98% accuracy on 645 unseen test images. As no product-to-packaging outcome labels exist, no recommendation accuracy is reported.

B. Future Work

Planned work includes exactly matching SKU-level products and improving catalogue matching beyond category-level detection. This includes automatic weight measurement using an IoT load-cell sensor, improved environmental estimates from reliable per-package mass data, and a larger packaging catalogue. We would test it on real-world camera images, collect seller packaging decisions and their outcomes (which would allow a learning-based ranking model), and improve handling of images that contain several objects.

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